EE 5239 Nonlinear Optimization

Lecture 3: First Order Methods

Mingyi Hong

University Of Minnesota

Last Time

- Convergence analysis of general descent method
- Works for general unconstrained problem, not necessarily convex problems
- Stepsize selection is important to guarantee convergence
  1. For constant stepsize steepest descent, stepsize should be less than $(2-\epsilon)/L$
  2. For other gradient type methods (with $\nabla f(x)$ being Lipschitz continuous with constant L)
     1. There exits a scalar ϵ such that for all r
        $$\epsilon < \alpha_r \le -\frac{(2-\epsilon)\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r\rangle}{L\|\mathbf{d}^r\|^2}$$
     2. $\alpha_r \to 0$, and $\sum_{r=1}^{\infty} \alpha_r = \infty$ (i.e., $\alpha_r = \frac{1}{r}$)

Last Time

- We also talked about the rate of convergence for strongly convex problems (and saw examples)
- When applying steepest gradient descent method, **global** linear convergence can be achieved for strongly convex problems
- The rate is dependent on the "condition number" of the problem

Last Time

- Key steps in proving convergence for gradient descent
  1. Bounding the descent with the size of the gradient
     $$f(\mathbf{x}^{r+1}) \le f(\mathbf{x}^r) - \frac{L}{2}\|\nabla f(\mathbf{x}^r)\|^2$$
  2. Or bounding the descent with the size of
     $$\langle \nabla f(\mathbf{x}^r), \mathbf{d}^r \rangle$$
- Key steps in proving rate of convergence
  1. S1: Sufficient Descent (the decrease achieved after each iteration):
     $$f(\mathbf{x}^{r+1}) - f(\mathbf{x}^r) \le \ldots$$
  2. S2: Estimating cost-to-go (how far away we are from the optimal solution):
     $$f(\mathbf{x}^r) - f(\mathbf{x}^*) \le \ldots$$

Comments

- Although simple, and appears to be naive, the simplest GD method still attracts lots of attentions.
- We can ask the following question: Does gradient descent converges to points beyond stationary solutions?
- Active research areas, recent papers include (and the references therein, and those papers that cite it)
  Lee et al 2016, "Gradient Descent Converges to Minimizers", COLT 2016
- **Why convergence analysis?**

Outline

1. Review
2. Motivating Examples
3. The Scaled Gradient Methods
4. Other First Order Methods
   - Coordinate Descent Method
   - Incremental Methods
   - Conjugate Direction Methods

This Time

- Introduce a few more efficient first order methods
- Including incremental methods, coordinate descent, etc
- Applications, theoretical analysis, etc

Example 1: Coordinate Descent

- As always, start with examples...
- What is gradient descent

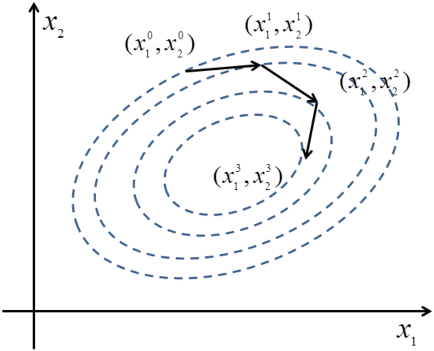

Figure 0.1: Illustration of gradient descent

Example 1: Coordinate Descent

- What is coordinate descent (CD)

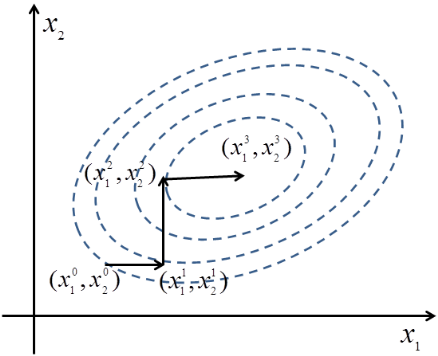

Figure 0.2: Illustration of coordinate descent

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More on Scaled Steepest Descent

- The section on "Scaling and Steepest Descent" in the book
- Recall that in gradient descent algorithm, convergence rate is dependent on the "condition number"
- What can we do if we have a problem that is ill conditioned?
  1. Scale the data directly (as in the housing price example)
  2. Scale the algorithm properly...

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- Recall that in gradient descent algorithm, convergence rate is dependent on the "condition number"
- What can we do if we have a problem that is ill conditioned?
  1. Scale the data directly (as in the housing price example)
  2. Scale the algorithm properly...
- Consider the following more general scaled steepest descent

$$\mathbf{x}^{r+1} = \mathbf{x}^r - \alpha_r \mathbf{D}^r \nabla f(\mathbf{x}^r)$$

- $\mathbf{D}^r \succ 0$ is some positive definite matrix; common choice, a diagonal matrix: $\mathbf{D}^r = \text{diag}[d_1, d_2, \cdots, d_n]$

More on Scaled Steepest Descent

- For simplicity, suppose $\mathbf{D}^r = \mathbf{D}$ (constant scaling for all iterations)

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$$\mathbf{x}^{r+1} = \mathbf{x}^r - \alpha_r \mathbf{D} \nabla f(\mathbf{x}^r)$$

- Consider the change of variable $\mathbf{x} = \mathbf{S}\mathbf{y}$ with $\mathbf{S} = (\mathbf{D})^{1/2}$
- Then in the space of $\mathbf{y}$, the problem becomes

$$\begin{aligned} \text{minimize} \quad & h(\mathbf{y}) = f(\mathbf{S}\mathbf{y}) \\ \text{subject to} \quad & \mathbf{y} \in \mathbb{R}^n \end{aligned}$$

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- Apply steepest descent to this problem

$$\begin{aligned} \mathbf{y}^{r+1} &= \mathbf{y}^r - \alpha_r \nabla h(\mathbf{y}^r) \\ \text{multiply } \mathbf{S} \text{ on both sides } \mathbf{S}\mathbf{y}^{r+1} &= \mathbf{S}\mathbf{y}^r - \alpha_r \mathbf{S} \nabla h(\mathbf{y}^r) \\ \text{equivalent to } \mathbf{x}^{r+1} &= \mathbf{x}^r - \alpha_r \mathbf{D} \nabla f(\mathbf{x}^r) \end{aligned}$$

More on Scaled Steepest Descent

- Performing scaled steepest descent on $\mathbf{x}$, is precisely equivalent to performing the unscaled steepest descent on $\mathbf{y}$

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- Performing scaled steepest descent on $\mathbf{x}$, is precisely equivalent to performing the unscaled steepest descent on $\mathbf{y}$
- **Any benefit for this observation?**

More on Scaled Steepest Descent

- Then the conclusion we had from last time also holds, that is for strongly convex problems, the convergence rate will be

$$e(\mathbf{y}^{r+1}) \leq (1 - \frac{\sigma}{L})e(\mathbf{y}^r) := \beta e(\mathbf{y}^r)$$

- Linear convergence, with constant $\beta = (1 - \frac{\sigma}{L})$
- **Note**: σ and L is the smallest and largest eigenvalue of $\nabla^2 h(\mathbf{y})$!

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- **Note**: σ and L is the smallest and largest eigenvalue of $\nabla^2 h(\mathbf{y})$!

- We have

$$\nabla^2 h(\mathbf{y}) = \mathbf{S} \nabla^2 f(\mathbf{x}) \mathbf{S}$$

- This means that we can pick the $\mathbf{S}$ to make σ and L close!
- We should pick $\mathbf{S}$ as close as $(\nabla^2 f(\mathbf{x}))^{-1/2}$, or pick the direction $\mathbf{D}^r$ as close as $(\nabla^2 f(\mathbf{x}))^{-1}$!

Outline

Coordinate Descent Method

- Consider $\mathbf{x} = (x_1, \cdots, x_n)$, and

$$\min f(x_1, x_2, \cdots, x_n)$$

- Choose the search directions along the coordinate directions
- Iterate through the list of search directions cyclically

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- Choose the search directions along the coordinate directions
- Iterate through the list of search directions cyclically
- **Version 1** (exact minimization): at iteration $r+1$, pick $i = rem(r, n) + 1$

$$\mathbf{x}_i^{r+1} = \arg\min_{x_i} f(x_1^r, \cdots, x_{i-1}^r, x_i, x_{i+1}^r, \cdots, x_n^r)$$
$$\mathbf{x}_j^{r+1} = \mathbf{x}_j^r, \ \forall\ j \neq i$$

where $rem(r, n)$ is the remainder of r/n.

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- **Version 2** (gradient step): at iteration $r+1$, pick $i = rem(r, n) + 1$

$$\mathbf{x}_i^{r+1} = \mathbf{x}_i^r - \alpha_i^r \nabla_i f(x^r)$$
$$\mathbf{x}_j^{r+1} = \mathbf{x}_j^r, \ \forall\ j \neq i$$

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Coordinate Descent Method

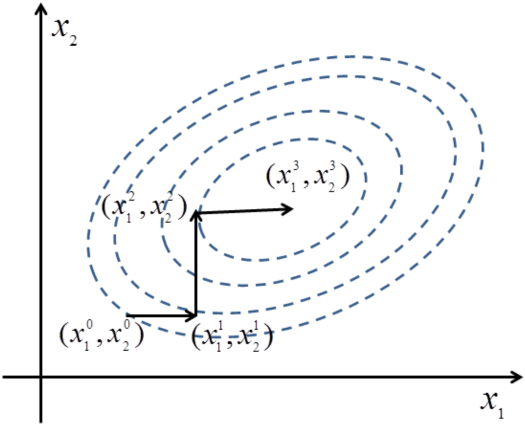

Figure 1.1: Illustration of Coordinate Descent Method

Coordinate Descent Method

- What is the benefit?
- Each step very small dimension, simple to perform
- Possible to avoid expensive operations such as matrix inversion
- Can be parallelized (to be discussed later)
- Convergence to first-order optimal solution easy to argue

Example of Coordinate Descent

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$$\min_{\mathbf{x}} \frac{1}{2} \left\| \mathbf{A}\mathbf{x} - \mathbf{b} \right\|^2$$

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- How to do coordinate descent?

$$\min_{\mathbf{x}} \frac{1}{2} \left\| \sum_{i=1}^{n} \mathbf{A}_i x_i - \mathbf{b} \right\|^2$$

where $\mathbf{A}_i$ represents ith column of $\mathbf{A}$, x_i is the ith element of $\mathbf{x}$

- Minimize with the ith coordinate:

$$\min_{x_i} \frac{1}{2} \left\| \mathbf{A}_i x_i + \sum_{j \neq i}^{n} \mathbf{A}_j x_j^r - \mathbf{b} \right\|^2$$

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- Solution very simple (no inversion needed!)

$$x_i^{r+1} = \frac{1}{\mathbf{A}_i^T \mathbf{A}_i} \mathbf{A}_i^T (\mathbf{b} - \sum_{j \neq i}^{n} \mathbf{A}_j x_j^r)$$

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Incremental Gradient Methods

- Let's again consider the least square problem (with m data points)

$$\min \quad f(\mathbf{x}) = \frac{1}{2}\|\mathbf{A}\mathbf{x} - \mathbf{b}\|^2 = \frac{1}{2}\sum_{i=1}^{m}\|\mathbf{A}^i\mathbf{x} - \mathbf{b}^i\|^2 := \frac{1}{2}\sum_{i=1}^{m} g_i(\mathbf{x})$$

  $\mathbf{A}^i$, $\mathbf{b}^i$ represents the ith row of $\mathbf{A}$ and $\mathbf{b}$, or the ith piece of data

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- The gradient method needs **all data** (or all $g_i's$) in each iteration
- Consider the following incremental method: At iteration $r+1$

Let $\psi_0 = \mathbf{x}^r$

Inner Loop $\psi_i = \psi_{i-1} - \underbrace{\alpha_r \nabla g_i(\psi_{i-1})}_{\text{a single data point}}, \; i = 1, \cdots, m$

Update the variable $\mathbf{x}^{r+1} = \psi_m$

- Total m inner loops; What is the advantage of incrementalism?

View as Gradient Method with Errors

- View the incremental method as gradient method with errors

$$\mathbf{x}^{r+1} = \underbrace{\mathbf{x}^r - \alpha_r \sum_{i=1}^{m} \nabla g_i(\mathbf{x}^r)}_{\text{The usual gradient step}} \underbrace{+ \alpha_r \sum_{i=1}^{m} (\nabla g_i(\mathbf{x}^r) - \nabla g_i(\psi_{i-1}))}_{\text{The error term}}$$

- Error term proportional to stepsize α_r

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- Error term proportional to stepsize α_r
- Convergence or a diminishing stepsize: square summable, infinite travel

$$\alpha_r \to 0, \quad \underbrace{\sum_r \alpha_r = \infty}_{\text{infinite travel}}, \quad \underbrace{\sum_r \alpha_r^2 < \infty}_{\text{square summable}} \tag{2.1}$$

- Convergence to a neighborhood of $\mathbf{x}^*$ for a constant stepsize

Comments

- Incremental type of algorithm is an old algorithm; see the following for a survey
  "Incremental Gradient, Subgradient, and Proximal Methods for Convex Optimization: A Survey", D. P. Bertsekas 2010.

- Recently it has attracted significant attention in ML and optimization communities

- Significant progress has been made to develop variants of incremental algorithm that achieves linear convergence

- Closely related to the stochastic gradient descent (SGD) algorithm

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Conjugate Direction Methods

- Aim to improve convergence rate of steepest descent, without incurring the complexity of Newton's method
- For quadratic problems, it requires n iterations to minimize $f(x) = \frac{1}{2}x^T Qx - b^T x$ with Q an $n \times n$ **positive definite** matrix.
- Converges in **finite steps** (between GD and Newton)

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- Materials in Section 2.1 in (Ver 3), and Section 1.6 (Ver 2)
- **Readings.**
  1. For **Version 3**. Page 120-125 before Proposition 2.1.1, and know how to implement preconditioned method Eq. (2.29) –(2.31)
  2. For **Version 2**. Section 1.6 before Proposition 1.6.1, and know how to implement preconditioned method Eq. (1.163) –(1.168)

Conjugate Direction Methods

- Direction $d^0, \dots, d^r$ are Q-conjugate, if

$$(d^i)^T Q d^j = 0, \; \forall \; i \neq j.$$

- Conjugate directions are linearly independent (proof in book)
- Example [see figure in book]

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- Generic conjugate direction method:

$$x^{r+1} = x^r + \alpha_r d^r$$

where

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2. α_r is obtained by line minimization [closed-form solution?]

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- Let us analyze the convergence of CG method ([suppose d^r's are available])

Convergence of CG

- Show that for all $r \geq 0$, each $g^{r+1} := \nabla f(x^{r+1})$ generated up to termination is **orthogonal** to each direction $(d^0, d^1, \dots, d^r)$.
  - $\star$ For each $i \geq 0$, exact line search implies
  $$\frac{\partial f(x^i + \alpha d^i)}{\partial \alpha}\Big|_{\alpha=\alpha^i} = (\nabla f(x^{i+1}))^T d^i = 0. \qquad (\star)$$
  - $\star$ Moreover, for any $i < r$, we have

$$\begin{aligned}
(\nabla f(x^{r+1}))^T d^i &= (Qx^{r+1} - b)^T d^i \\
&= \left(x^{i+1} + \sum_{j=i+1}^{r} \alpha_j d^j\right)^T Qd^i - b^T d^i \\
&= (x^{i+1})^T Q d^i - b^T d^i \\
&= \left(\nabla f(x^{i+1})\right)^T d^i = 0
\end{aligned} \tag{3.1}$$

The last equation is from $(\star)$

Convergence of CG

- At x_0, we are allowed to go to the directions spanned by $(d^0, d^1, \cdots, d^r)$:

$$x^0 + \beta_0 d^0 + \beta_1 d^1 + \cdots + \beta_r d^r$$

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- **Question**: The best solution computable by changing β_i's?
- Let $\beta = (\beta_0, \beta_1, \ldots, \beta_r)^T$ and $\alpha = (\alpha_0, \alpha_1, \ldots, \alpha_r)^T$. Then

$$\left.\frac{\partial f(x^0+\beta_0 d^0+\beta_1 d^1+\cdots+\beta_r d^r)}{\partial \beta_i}\right|_{\beta=\alpha} = (\nabla f(x^{r+1}))^T d^i = 0.$$

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- Therefore, we have

$$x^{r+1} = \arg\min_{x \in \mathcal{M}^r} f(x),$$

$$\text{where } \ \mathcal{M}^r = \{x \mid x = x^0 + v,\ v \in \mathsf{span}(d^0, d^1, \ldots, d^r)\}$$

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- Therefore, we have

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$$\text{where } \ \mathcal{M}^r = \{x \mid x = x^0 + v, \ v \in \mathsf{span}(d^0, d^1, \ldots, d^r)\}$$

- This further implies $f(x^r) \downarrow f(x^*)$ monotonically, and x^n minimizes $f(x)$ over $\mathbb{R}^n$.

Generating Conjugate Directions

- Given set of linearly independent vectors $\xi^0, \ldots, \xi^r$, we can construct a set of Q-conjugate directions $d^0, \ldots, d^r$ as follows

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- Gram-Schmidt procedure. Start with $d^0 = \xi^0$. If for some $i < r$, $d^0, \ldots, d^i$ are Q-conjugate; We will generate d^{i+1} as follows

$$d^{i+1} = \xi^{i+1} + \sum_{m=0}^{i} c_{(i+1),m} d^m$$

choose $c_{(i+1),m}$ so d^{i+1} is Q-conjugate to $d^0, \ldots, d^i$.

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  choose $c_{(i+1),m}$ so d^{i+1} is Q-conjugate to $d^0, \ldots, d^i$.

- Let $j < i$, multiple Qd^j to the right hand side we obtain:

$$(d^{i+1})^T Q d^j = (\xi^{i+1})^T Q d^j + \left(\sum_{m=0}^{i} c_{(i+1),m} d^m \right)^T Q d^j = 0$$

  implying (note that d^0, d^i are Q-conjugate)

$$c_{(i+1),j} = -\frac{(\xi^{i+1})^T Q d^j}{(d^j)^T Q d^j}, \quad \forall 0 \leq j \leq i$$

The Conjugate Gradient Methods

- The conjugate gradient (CG) method is the conjugate direction method where the direction is generated using gradients

- Apply Gram-Schmidt to the vectors $\xi^r = -g^r = -\nabla f(x^r)$, $r = 1, \ldots, n-1$

$$d^r = -g^r + \sum_{j=0}^{r-1} \frac{(g^r)^T Q d^j}{(d^j)^T Q d^j} d^j$$

- Furthermore, the method terminates with an optimal solution after at most n steps.

Conjugate Gradient Method: Summary

- The algorithm can be summarized as follows

$$g^0 = \nabla f(x^0), \quad d^0 = -g^0$$

$$d^r = -g^r + \sum_{j=0}^{r-1} \frac{(g^r)^T Q d^j}{(d^j)^T Q d^j} d^j, \; r = 1, \cdots, n-1$$

$$g^r = \nabla f(x^r) = Qx^r - b$$

$$\alpha^r = \frac{(d^r)^T (b - Qx^r)}{(d^r)^T Q d^r} \qquad \text{[by the OPT condition of line search]}$$

$$x^{r+1} = x^r + \alpha_r d^r$$

Conclusion

- Today we introduced a few other first order methods
- Both incremental and coordinate descent methods can be viewed as taking "divide and conquer" approach